# Supplementary Material: Studying the Effects of Robot Distraction on School Shooter Behavior Using Virtual Reality

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This document provides supplementary technical details for the IJCAI-ECAI 2026 paper. It is hosted externally and is not part of the official proceedings version.

## A Shooter Prediction Model

Figure 1 provides an overview of the sequence-to-sequence model used to predict shooter motion. Starting on the left, the model receives a sequence of past shooter velocities, represented as two-dimensional components over the previous $n_{prev}$ timesteps. In parallel, the model receives spatial context inputs describing the surrounding environment, including occupancy and object states derived from the school layout. Occupancy information is encoded using a $21 \times 21$ Cartesian grid, while object states (alive NPCs, dead NPCs, open doors, and closed doors) are encoded by $20 \times 20$ polar grids centered on the shooter. These grids are flattened and stacked across time, producing input sequences of shape $(n_{prev}, 441)$ for occupancy and $(n_{prev}, 400)$ for each object channel.

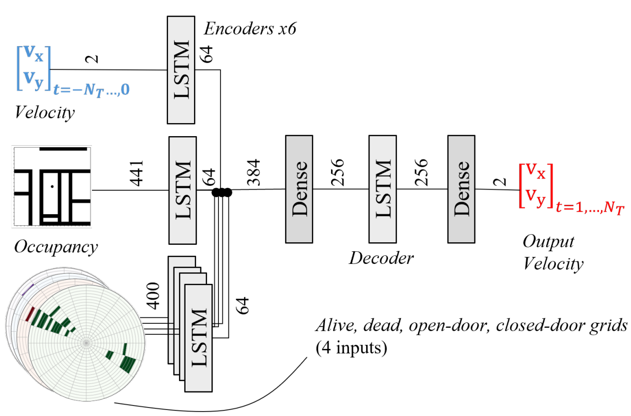


Figure 1: An overview of our shooter trajectory model. Previous trajectory and object maps are encoded by an LSTM layer. The final states of the encoder are decoded, concatenated with the context vector, and passed through a dense layer for trajectory prediction.

Each input stream is processed independently by an LSTM encoder with 64 cells, producing a fixed-length latent representation for each modality. The resulting latent vectors are concatenated and passed through a fully connected dense layer, yielding a fused latent embedding of length 256. This embedding is then provided to an LSTM decoder, which generates future predictions one timestep at a time, with each prediction conditioned on the previous decoder state. Finally, a dense output layer maps each decoder state to a two-dimensional velocity prediction, producing the estimated future shooter velocity sequence shown on the right side of the figure.

## B Manipulation Checks

Manipulation checks are used to verify that study participants experienced different conditions as predicted. For our study each subject experienced a single approach–distraction combination. Measures of victim counts and gaze at the robot indicate that our approach/distraction manipulations did influence that dependent variable. We also conducted post-experiment surveys to capture each participant's subjective impression of the robot. Because participants had no within-subject comparison across approach styles or distraction intensities, subjective ratings primarily reflect absolute impressions rather than relative sensitivity to the manipulations. Consistent with this, perceived robot aggressiveness clustered near the midpoint of the scale ($M = 3.2$) and showed only a small, non-significant positive association with approach condition ($r = .07$, $p = .37$). Perceived distraction was not directly measured, as distraction was examined behaviorally through gaze allocation and shooting behavior. These subjective post-experiment ratings should be interpreted as secondary to behavioral measures of manipulation effectiveness rather than as direct manipulation checks.

## C Statistical Modeling

For gaze and victim count, participant-level heterogeneity varied little within subjects. With respect to gaze, for example, once the robot is noticed, participants fixated on it. We thus used GLMs. For victim counts, an initial Poisson regression revealed evidence of over-dispersion. A Negative Binomial model provided a superior fit (based on AIC) and was used for inference. For gaze, a least-squares model provided a better fit than a Gamma model. These models verified our prior analysis and showed gaze in the medium condition differed significantly from gaze in the high condition, providing additional support for H3.

### C.1 Mean Victim Count Model

**Descriptive Statistics**. Mean victim counts varied across conditions. The no robot baseline produced the highest mean number of victims, while conditions involving robot intervention—particularly those with aggressive, high-distraction behaviors—showed substantial reductions.

**Model Fit and Overdispersion.** An initial Poisson regression revealed evidence of overdispersion (residual deviance$/df > 1$), indicating greater variance in victim counts than expected under a Poisson process. A Negative Binomial (NB) model provided a superior fit ($AIC_{NB} < AIC_{Poisson}$) and was therefore used for subsequent inference.

**Main Effects of Condition.** The NB regression revealed a statistically significant main effect of condition on the number of victims, $\chi^2(df = 4, N = 150) \approx 60.519, p < .001$. The predicted mean victim counts from the NB model are provided in Table 1.

**Pairwise Comparisons.** Pairwise contrasts among conditions were computed on the incidence rate ratio (IRR) scale. Relative to the no robot baseline, the aggressive–high condition showed a significant reduction in expected victim count ($IRR = 0.535, 95\%$ *CI* $[0.319–0.896], p < .01$), corresponding to an approximate 46.5% decrease in fatalities. Comparing the aggressive-high to the aggressive–low condition also yielded a meaningful reduction ($IRR = 0.587, p < .05$), while the passive–high condition did not differ significantly from the baseline ($IRR \approx 1.4, p > .10$).

**Interpretation.** Together, these results demonstrate that the robot's behavioral aggressiveness and multi-modal distraction intensity significantly influenced its ability to mitigate casualties. Robots employing active, high-distraction tactics were most effective in reducing victim counts.

Table 1: Condition Predicted Mean Victims.

| Condition | Predicted Mean Victims |
|---|---|
| No-robot | 35.23 |
| Aggressive–Low | 32.10 |
| Aggressive–Med | 29.23 |
| Aggressive–High | 18.83 |
| Passive–High | 24.80 |

**Model Details.** The details for the mean victim model are provided in Figure 2.

### C.2 Robot Gaze Model

**Descriptive Statistics**. We next examined participants' visual attention toward the robot, measured as the percentage of time their gaze was directed at the robot (gazeAtRobot%). A linear model with condition as a fixed effect revealed a robust effect of condition on gaze allocation. Mean gaze percentages were 0.0% in the no robot baseline, 3.5% in the passive–high condition, 9.6% in aggressive–low, 10.1% in aggressive–high, and 12.9% in aggressive–medium.

**Model Fit.** A linear model of the form:

$$\text{gazeAtRobot\%} = \beta_0 + \beta_1 \, \text{condition} + \varepsilon$$

showed a strong effect of condition on gaze. In the ordinary least squares model (OLS) output, all non-zero-gaze conditions differ very sharply from the no robot baseline (p-values effectively $< 10^{-14}$), and there are clear, significant differences among the robot conditions themselves.

**Pairwise Comparisons.** Pairwise contrasts indicated that all robot conditions elicited substantially more gaze than the no-robot baseline (all $p < .002$), and that each aggressive condition produced more gaze than the passive–high

```
Full data Negative Binomial GLM:
                 Generalized Linear Model Regression Results
==============================================================================
Dep. Variable:                victims   No. Observations:                  150
Model:                            GLM   Df Residuals:                      145
Model Family:        NegativeBinomial   Df Model:                            4
Link Function:                    Log   Scale:                          1.0000
Method:                          IRLS   Log-Likelihood:                -649.35
Date:                Mon, 10 Nov 2025   Deviance:                       60.519
Time:                        20:24:25   Pearson chi2:                     37.8
No. Iterations:                     5   Pseudo R-squ. (CS):            0.04353
Covariance Type:            nonrobust
==============================================================================================
                                   coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------------------
Intercept                        2.9356      0.187     15.669      0.000       2.568       3.303
C(cond_short)[T.aggressive_low]  0.5332      0.264      2.023      0.043       0.017       1.050
C(cond_short)[T.aggressive_med]  0.4397      0.264      1.667      0.096      -0.077       0.957
C(cond_short)[T.no_robot]        0.6264      0.263      2.378      0.017       0.110       1.143
C(cond_short)[T.passive_high]    0.2752      0.264      1.042      0.297      -0.243       0.793
==============================================================================================
```

Figure 2: Details for the Negative Binomial (NB) model.

```
OLS (Full data):
                            OLS Regression Results
==============================================================================
Dep. Variable:       gazeAtRobot_pct   R-squared:                       0.575
Model:                           OLS   Adj. R-squared:                  0.564
Method:                Least Squares   F-statistic:                     49.13
Date:               Tue, 11 Nov 2025   Prob (F-statistic):           4.53e-26
Time:                       01:49:20   Log-Likelihood:                -423.28
No. Observations:                150   AIC:                             856.6
Df Residuals:                    145   BIC:                             871.6
Df Model:                          4
Covariance Type:           nonrobust
=====================================================================================================
                                       coef    std err          t      P>|t|      [0.025      0.975]
-----------------------------------------------------------------------------------------------------
Intercept                           10.0848      0.755     13.353      0.000       8.592      11.577
C(cond_short)[T.aggressive_low]     -0.5316      1.068     -0.498      0.619      -2.643       1.579
C(cond_short)[T.aggressive_med]      2.8134      1.068      2.634      0.009       0.702       4.924
C(cond_short)[T.no_robot]          -10.0848      1.068     -9.442      0.000     -12.196      -7.974
C(cond_short)[T.passive_high]       -6.6161      1.068     -6.195      0.000      -8.727      -4.505
==============================================================================
Omnibus:                      18.754   Durbin-Watson:                   2.205
Prob(Omnibus):                 0.000   Jarque-Bera (JB):               26.295
Skew:                          0.700   Prob(JB):                     1.95e-06
Kurtosis:                      4.499   Cond. No.                         5.83
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
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Figure 3: Details for the ordinary least squares model.

robot (differences of approximately 6–9 percentage points, all $p < 10^{-7}$). Within the aggressive configurations, the aggressive–medium condition attracted significantly more gaze than both aggressive–low and aggressive–high ($\approx 3.4$ and 2.8 percentage points more, respectively; $p \leq .009$), whereas aggressive–high and aggressive–low did not differ reliably from each other $(p = 0.62)$.

**Interpretation.** These results suggest that an intermediate level of multimodal distraction (aggressive–medium) was most effective at capturing and sustaining visual attention.

**Model Details.** The details for the robot gaze model are provided in Figure 3.

# D Preregistered Hypotheses

The preregistered hypotheses included the material below. In the main document, we refer to ”a robot that races” and ”a robot that follows” as aggressive and passive approach, respectively. Moreover, we refer to the ”no distractors,” ”lights only to distract,” and ”uses lights and smoke to distract” as low, medium, and high distraction levels, respectively. In total, there are 45 hypotheses for the experiment.

## D.1 Behavioral Objectives

In our between-subject manipulation of robot presence, approach strategy, and distractor type, we predict:

1. *[Directional]* As opposed to no robot, the presence of a robot will decrease the total number of shots fired.
2. *[Directional]* As opposed to no robot, the presence of a robot will decrease the count of victims.
3. *[Directional]* As opposed to no robot, the presence of a robot will decrease the distance traveled by the participant.
4. *[Directional]* As opposed to following, a robot that races the shooter to his next position (+5s) will decrease the total number of shots fired.
5. *[Directional]* As opposed to following, a robot that races the shooter to his next position (+5s) will decrease the count of victims.
6. *[Directional]* As opposed to following, a robot that races the shooter to his next position (+5s) will decrease the distance traveled by the participant.
7. *[Directional]* As opposed to following, a robot that races the shooter to his next position (+5s) will increase the percentage of shots fired at the robot.
8. *[Directional]* As opposed to following, a robot that races the shooter to his next position (+5s) will increase the percentage of time gazing at the robot.
9. *[Directional]* As opposed to no distractors, a robot that uses lights only to distract will decrease the total number of shots fired.
10. *[Directional]* As opposed to no distractors, a robot that uses lights only to distract will decrease the count of victims.

11. *[Directional]* As opposed to no distractors, a robot that uses lights only to distract will decrease the distance traveled by the participant.
12. *[Directional]* As opposed to no distractors, a robot that uses lights only to distract will increase the percentage of shots fired at the robot.
13. *[Directional]* As opposed to no distractors, a robot that uses lights only to distract will increase the percentage of time gazing at the robot.
14. *[Directional]* As opposed to using lights only to distract, a robot uses lights and smoke to distract will decrease the total number of shots fired.
15. *[Directional]* As opposed to using lights only to distract, a robot uses lights and smoke to distract will decrease the count of victims.
16. *[Directional]* As opposed to using lights only to distract, a robot uses lights and smoke to distract will decrease the distance traveled by the participant.
17. *[Directional]* As opposed to using lights only to distract, a robot uses lights and smoke to distract will increase the percentage of shots fired at the robot.
18. *[Directional]* As opposed to using lights only to distract, a robot uses lights and smoke to distract will increase the percentage of time gazing at the robot.
19. *[Directional]* As opposed to no distractors, a robot uses lights and smoke to distract will decrease the total number of shots fired.
20. *[Directional]* As opposed to no distractors, a robot uses lights and smoke to distract will decrease the count of victims.
21. *[Directional]* As opposed to no distractors, a robot uses lights and smoke to distract will decrease the distance traveled by the participant.
22. *[Directional]* As opposed to no distractors, a robot uses lights and smoke to distract will increase the percentage of shots fired at the robot.
23. *[Directional]* As opposed to no distractors, a robot uses lights and smoke to distract will increase the percentage of time gazing at the robot.

### D.2 Demographic Representativeness

In our between-subject manipulation of robot presence, approach strategy, and distractor type, we predict:

24. *[Non-directional]* With respect to distribution of sex, the test subject population follows the shooter population.
25. *[Non-directional]* With respect to distribution of sex, the test subject population differs from the general population.
26. *[Non-directional]* With respect to distribution of race, the test subject population differs from the shooter population.
27. *[Non-directional]* With respect to distribution of race, the test subject population differs from the general population.
28. *[Non-directional]* With respect to distribution of religion, the test subject population differs from the shooter population.
29. *[Non-directional]* With respect to distribution of religion, the test subject population follows the general population.
30. *[Non-directional]* With respect to distribution of having children, the test subject population follows the shooter population.
31. *[Non-directional]* With respect to distribution of having children, the test subject population follows the general population.
32. *[Non-directional]* With respect to distribution of marriage status, the test subject population follows the shooter population.
33. *[Non-directional]* With respect to distribution of marriage status, the test subject population differs from the general population.
34. *[Non-directional]* With respect to distribution of education level, the test subject population differs from the shooter population.
35. *[Non-directional]* With respect to distribution of education level, the test subject population follows the general population.

### D.3 Survey and Behavioral Outcomes

In our between-subject manipulation of robot presence, approach strategy, and distractor type, we predict:

36. *[Directional]* There is a positive correlation between participants feeling “immersed in the environment” and distance traveled.
37. *[Directional]* There is a negative correlation between participants feeling “immersed in the environment” and number of shots fired at non-robot entities.
38. *[Directional]* There is a positive correlation between participants feeling they “knew the task and took it seriously” and number of victims.
39. *[Directional]* There is a negative correlation between participants feeling they “people in the environment seemed realistic” and number of victims.
40. *[Directional]* There is a negative correlation between participants feeling they “people in the environment seemed realistic” and number of shots fired at non-robot entities.
41. *[Directional]* There is a negative correlation between participants feeling they “were able to go where I wanted in the environment” and perceiving the robot as “alive.”
42. *[Directional]* There is a negative correlation between participants feeling they “were able to go where I wanted in the environment” and perceiving the robot as “competent.”
43. *[Directional]* There is a negative correlation between participants feeling they “were able to go where I wanted in the environment” and perceiving the robot as “aggressive.”

44. *[Directional]* There is a positive correlation between participants perceiving the robot as "competent" and feeling "stalled / obstructed" by the robot.
45. *[Directional]* There is a positive correlation between participants perceiving the robot as "aggressive" and feeling "surprised" by the robot.

## E Additional Measures

In addition to the reported number of victims and gaze towards the robot, we also analyzed the effects of robot conditions on the shooter's distance traveled, number of shots fired, and portion of shots directed at the robot. The figures below give the results of these additional measures.

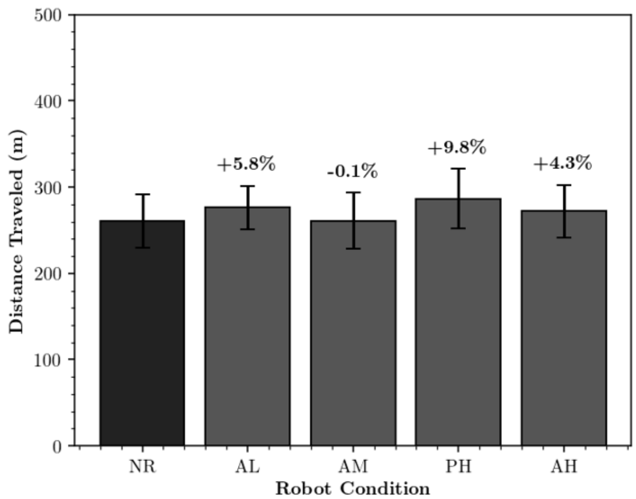


Figure 4: Shooter distance traveled per robot condition: no robot (NR), aggressive with low-distraction (AL), aggressive with medium-distraction (AM), passive with high-distraction (PH), and aggressive with high-distraction (AH). Error bars show 95% CI. None of the robot conditions were significantly different from the no-robot control.

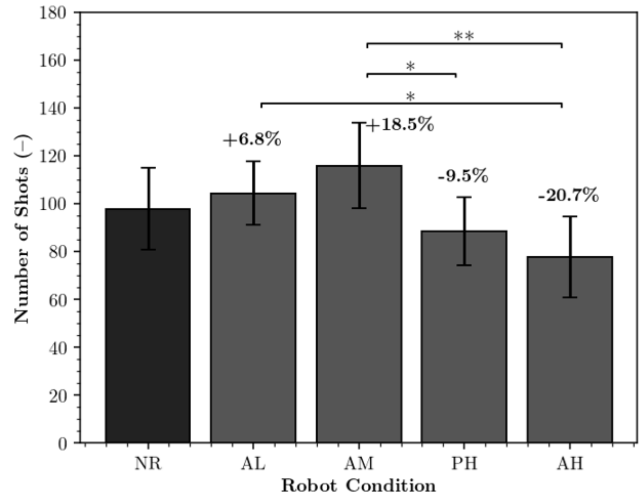


Figure 5: Number of shots fired per robot condition: no robot (NR), aggressive with low-distraction (AL), aggressive with medium-distraction (AM), passive with high-distraction (PH), and aggressive with high-distraction (AH). Error bars show 95% CI. Bold labels indicate differences from the no-robot control. Asterisks denote significance from Welch's $t$-test: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

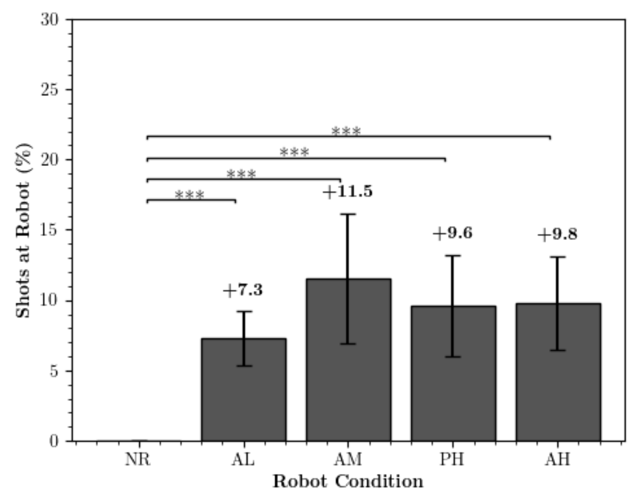


Figure 6: Portion of shots directed at the robot per robot condition: no robot (NR), aggressive with low-distraction (AL), aggressive with medium-distraction (AM), passive with high-distraction (PH), and aggressive with high-distraction (AH). Error bars show 95% CI. Bold labels indicate differences from the no-robot control. Asterisks denote significance from Welch's $t$-test: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.